# JURISGUARD: SECURE LEGAL DOCUMENT EXTRACTION AND ARCHIVING SYSTEM

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# CHAPTER 1

# INTRODUCTION

## Background of Study

In the global landscape of digital transformation, secure Document Management Systems (DMS) have become essential for maintaining operational efficiency, preserving document integrity, and strengthening resilience against increasing cybersecurity risks [1]. Current international directions in document security emphasize Zero Trust Architecture, where access to protected resources is continuously verified rather than assumed, because organizations that handle sensitive records remain vulnerable to unauthorized access, ransomware incidents, and internal data tampering when controls are weak [2]. In this context, secure legal and administrative recordkeeping is no longer only a technical concern but also an institutional requirement for trustworthy and accountable information management.

This concern also connects with broader development priorities. Effective and secure management of sensitive records supports Sustainable Development Goal (SDG) 16, particularly the promotion of effective, accountable, and transparent institutions, because information systems shape how institutions protect data, maintain traceability, and deliver reliable public and legal services. In the same way, the study is naturally aligned with the DNSC Research, Development, and Extension (RDE) agenda by advancing an ICT-based solution that improves institutional processes, strengthens data protection practices, and promotes socially responsive digital innovation in service-oriented environments.

In the Philippines, records digitization initiatives in State Universities and Colleges show growing recognition that paper-based recordkeeping is inefficient and vulnerable to loss, damage, and retrieval delays [3]. However, the National Privacy Commission warns that spreadsheets remain a weak storage medium for personal data because they commonly lack sufficient security controls, accountability mechanisms, and access restrictions [4]. Despite ongoing digitization efforts, institutions continue to face cybersecurity and service-delivery challenges that weaken compliance and expose sensitive records to accidental deletion, unauthorized copying, and other cyber risks [5], [6].

There is also limited integration between intelligent document processing and security-centered archival design in existing systems. Recent studies show that OCR and legal information extraction can transform scanned documents into searchable and structured digital records [8], [9], [11], while secure document management studies emphasize stronger governance, encryption, and access control in repositories handling sensitive information [1], [2], [12]. However, these approaches are often treated separately, leaving a practical gap for legal and administrative environments that need both accurate metadata extraction from physical documents and fortified archival protection within a single workflow.

At the local level, Philippine implementation studies also suggest that digitization efforts are shaped by infrastructure readiness, user capability, interoperability needs, and organizational support [3], [5], [7]. In legal and administrative settings similar to the present study context, sensitive physical documents such as IDs, affidavits, certificates, and court orders are still commonly handled through manual encoding into unsecured spreadsheets. This process is time-consuming during high-volume intake, makes retrieval less efficient, and leaves confidential client and case information vulnerable to loss, alteration, or unauthorized access.

Despite the relevance of prior studies, there remains limited research that unifies OCR-assisted legal document extraction, verification-based encoding, centralized archival storage, cryptographic protection, role-based access control, multi-factor authentication, and immutable audit logging in one legal-domain platform [1], [2], [8], [9], [11], [12], [13], [14], [16]. This unresolved gap provides the main justification for the study because existing works either emphasize document extraction without sufficiently strengthening archival accountability or prioritize repository security without addressing structured intake from physical legal documents.

This study aims to develop JurisGuard, an intelligent legal document extraction and cryptographically secured archiving system for legal case information management. Specifically, it seeks to automate the extraction of case and client metadata from physical documents, support verification-based encoding, and provide a centralized repository protected through encryption, RBAC, MFA, and immutable audit logging. By doing so, the study intends to improve administrative workflow efficiency while strengthening compliance, accountability, and protection of sensitive legal records.

**Objectives of the Study**

The primary goal of this project is to develop JurisGuard, an intelligent legal document extraction and cryptographically secured archiving system that protects sensitive case data against threats while ensuring regulatory compliance. Specifically, the project aims:

1. To establish a centralized Digital Case Archive that will replace inefficient and unsecured Microsoft Excel spreadsheets for managing sensitive client and case information.

2. To develop an OCR-assisted module for extracting case and client metadata from physical legal documents into predefined digital fields.
3. To design and integrate a verification interface that allows manual validation of extracted data and supports the tracking, searching, and updating of case histories from intake to termination.
4. To implement multi-layered cybersecurity mechanisms, including cryptographic protection, role-based access control, multi-factor authentication, and immutable audit logging, to safeguard encoded data and ensure accountability.
5. To produce a secure and efficient legal case information management platform that enhances data retrieval, strengthens administrative workflow, and supports compliance with the Data Privacy Act of 2012.

**Significance of Study**

The development of JurisGuard introduces a proactive, security-first approach to legal data management. It transforms traditional, vulnerable manual encoding practices into a fortified digital ecosystem. The system also introduces a structured OCR-assisted approach to legal document processing by transforming physical records into searchable, structured data while maintaining a verification-based process that enhances operational efficiency and data integrity. The following are the study's main beneficiaries:

**Legal Organizations and Staff.** The system will minimize the risk of data breaches and ransomware attacks, ensuring that sensitive client information remains confidential and compliant with national privacy laws.

**Administrative and Legal Staff.** By replacing manual Excel entries with a centralized platform, the system reduces repetitive clerical work, shortens the time spent on data encoding and retrieval, supports faster client intake processing, and enables staff to focus more on core legal tasks, legal analysis, and client assistance while maintaining a clear audit trail.

**Clients.** JurisGuard ensures that the highly sensitive documents of clients, such as IDs and court orders, are stored securely, protecting them from unauthorized access and identity theft.

**Future Researchers and Developers.** This study provides a foundation for future research into Zero Trust legal repositories, OCR-assisted legal document extraction, and the integration of cybersecurity frameworks in small-scale administrative environments.

## Scope and Limitation

This study focuses on the design, development, and evaluation of JurisGuard, an intelligent legal document extraction and cryptographically secured archiving system specifically for case information management. The methodology involves the digitization of physical intake documents (IDs, affidavits, certificates, and court orders) into a secured database environment. The system is designed to process printed English and Filipino text from physical legal documents and to automate the extraction of case and client metadata into predefined fields through OCR-assisted processing and verification. The system will handle the complete Information Lifecycle, including data entry, encryption at rest, role-based access, and eventual case termination. Technical features include the implementation of MFA, RBAC, a backend audit logging mechanism, and a verification interface for manual data validation and quality assurance.

The study is limited to the management of digital text and demographic data encoded from physical documents; it does not include the physical storage of the original paper documents. The platform is designed for a specific institutional or legal clinic setting and may require further customization for large-scale corporate firms. The system is optimized for printed documentation, with handwritten or highly stylized text intended for manual entry through the provided verification interface. While the system provides robust defense against common cyber threats like ransomware, its effectiveness is partially dependent on the users' adherence to security protocols, such as not sharing login credentials. The evaluation will focus on the system's security features, workflow efficiency, OCR-assisted extraction performance, and compliance with the Data Privacy Act of 2012.

## Review of Related Literature and Works

### Related Literature

The present review examines studies relevant to intelligent legal document extraction and cryptographically secured archiving systems. To improve clarity and organization, the discussion is grouped into major themes: secure document management and zero trust architectures, digitization and records governance in the Philippine context, OCR-based digitization and legal information extraction, identity and access control, and the remaining integration gap addressed by JurisGuard.

*Secure Document Management and Zero Trust Architectures*

Secure document management is now treated not only as a matter of storage efficiency but also as a matter of confidentiality, integrity, and accountability across the document lifecycle.

Muñoz-Hernandez, Morales-Sandoval, and Garcia-Hernandez [1] argued that a robust secure document management system must combine controlled access, traceability, and end-to-end protection of digital records. In the same direction, Miloslavskaya and Tolstoy [2] applied Zero Trust Architecture to document management environments and emphasized that access to protected resources should be continuously verified rather than implicitly trusted. This security orientation is reinforced by NIST SP 800-207, which defines zero trust as a model that continuously verifies identities, devices, and access requests before resources are granted [12]. For systems such as JurisGuard that will manage sensitive legal records, these works establish the importance of encryption, strict authorization, and resource-centered protection.

*Digitization and Records Governance in the Philippine Context*

Recent Philippine studies also show that digitization efforts are most effective when paired with governance and security controls. Pardiñan, Bondad, and Mangubat [3] found that paper-based records management in Philippine State Universities and Colleges remains vulnerable to misplacement, damage, and retrieval inefficiency, which strengthens the case for digitized records platforms. The National Privacy Commission likewise warned that spreadsheets are a weak storage medium for personal data because they often lack adequate security controls, accountability mechanisms, and access restrictions [4]. Related local studies on digital service delivery and cybersecurity further show that Philippine institutions continue to face infrastructure limitations, inconsistent technical readiness, and security program gaps that can weaken compliance and expose sensitive records to risk [5], [6]. Even in implementation-oriented sectors, such as Philippine primary care settings, digital transformation has been shown to depend on user readiness, process redesign, and institutional support [7]. Collectively, these works show that digitization alone is insufficient unless it is paired with stronger security and governance mechanisms.

*OCR-Based Digitization and Legal Information Extraction*

A second theme in the literature concerns the transformation of scanned or image-based documents into structured and usable data. Premasiri et al. [8] surveyed legal information extraction research and showed that legal-domain extraction increasingly depends on specialized methods for identifying entities, events, arguments, and relationships from complex documents. Castano et al. [9] further demonstrated that context-aware legal information extraction can improve the accuracy and usefulness of structured legal data. In the Philippine legal domain, Virtucio et al. [10] showed that natural language processing and machine learning can be applied effectively to legal datasets, indicating that computational approaches are viable for legal information systems. At the archival level, Jayoma, Moyon, and Morales [11] demonstrated that OCR-based document archiving and indexing can improve storage and retrieval by converting scanned documents into searchable records. Together, these studies suggest that legal records can move beyond plain storage and can instead become structured, searchable, and analysis-ready information assets.

*Identity, Access Control, and Audit-Centered Accountability*

Strong identity assurance and traceable access controls are also recurring requirements in the literature on secure repositories. NIST Digital Identity Guidelines [13] recommend authentication mechanisms that are matched to the sensitivity and risk level of the protected service, which supports the use of multi-factor authentication for systems storing legal records. Maulina and Rasjid [14] likewise argued that digital evidence repositories require fine-grained access control by integrating role-based and attribute-based approaches so that users access only the information necessary for their responsibilities. At the implementation level, the OWASP Logging Cheat Sheet [16] stresses that application logs should record critical events such as access to sensitive data, privilege changes, data modifications, exports, and other security-relevant actions, and that such logs should be protected from tampering and unauthorized deletion. These recommendations directly support the inclusion of RBAC, MFA, and immutable audit logging in JurisGuard.

*Convergence of Intelligent Processing and Secure Archival Design*

Recent systems research points toward the growing convergence of intelligent document handling and strong archival protection. Han and Son [15] proposed a decentralized document management architecture that improves security and integrity through encryption, distributed permissions, blockchain-supported validation, and resilient repository design. Although their application domain is broader than legal case handling, the underlying principle is highly relevant: modern repositories must support both efficient document processing and verifiable security enforcement. This same convergence is implied across the reviewed studies, where extraction-oriented literature [8], [9], [11] and security-oriented literature [1], [2], [12] increasingly point toward integrated, rather than isolated, design approaches.

*Synthesis and Research Gap*

The reviewed literature reveals several consistent points. First, organizations are moving away from fragmented paper-based and spreadsheet-dependent recordkeeping toward centralized digital repositories with stronger governance controls [1], [3], [4]. Second, OCR and legal information extraction methods have been shown to improve indexing, retrieval, and the structuring of document content for downstream use [8], [9], [11]. Third, zero trust principles, strong authentication, access-control models, and tamper-aware logging are increasingly treated as foundational requirements for systems that manage sensitive records [2], [12], [13], [14], [16]. Despite these advances, there remains limited research that unifies OCR-assisted legal document extraction, verification-based data encoding, centralized archival storage, cryptographic protection, role-based access control, multi-factor authentication, and immutable audit logging in one platform tailored to legal case information management. This gap justifies the development of JurisGuard, which seeks to integrate these dimensions into a single secure workflow for legal records handling.

**Related Works**

Various document management and archival solutions have been developed to address digitization, information extraction, search, and security concerns in administrative and legal environments. Some systems emphasize OCR or legal information extraction, while others focus on access control, integrity, or decentralized storage. The reviewed studies show meaningful progress, but they also reveal that many existing solutions still address only one side of the problem, either intelligent extraction or strong repository protection, rather than both within a single legal records platform.

| **System** | **OCR / Extraction** | **Searchable Repository** | **RBAC / Access Control** | **MFA** | **Audit Logging** | **Encryption / Secure Storage** | **Verification Workflow** |
|---|---|---|---|---|---|---|---|
| ASKE Approach | ✓ | | | | | | |
| DSWD OCR Archive | ✓ | ✓ | | | | | |
| Records Digitization Platform | | ✓ | | | | | |
| Digital Evidence Access Model | | ✓ | ✓ | | ✓ | ✓ | |
| Decentralized DMS | | ✓ | ✓ | | ✓ | ✓ | |
| JurisGuard (Proposed) | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

***Table 1.*** *Comparison of related studies/systems and the proposed JurisGuard system.*

As shown in Table 1, existing studies and systems tend to emphasize either information extraction or secure archival protection, but seldom combine both at the level needed for legal case management. The ASKE legal information extraction approach [9] and the DSWD OCR-based archiving system [11] demonstrate the value of extraction and searchable repositories, yet they do not foreground advanced cybersecurity controls such as multi-factor authentication, cryptographic protection, and immutable logging. Philippine digitization studies [3] highlight the importance of workflow modernization, but they are not specifically framed as OCR-driven legal records platforms. By contrast, the unified access management model for digital evidence storage [14] and the decentralized document management architecture [15] emphasize access governance, integrity, and secure storage, but they do not focus on OCR-assisted intake of physical legal documents. JurisGuard distinguishes itself by integrating OCR-assisted extraction, verification-based encoding, centralized storage, role-based access control, multi-factor authentication, audit logging, and secure archival protection in one legal-domain system.

**Definition of Terms**

**Access Control.** Refers to the security mechanism that regulates which users can view, encode, modify, or manage case records and system resources within JurisGuard.

**Administrative Workflow**. Refers to the sequence of processes involved in encoding, retrieving, updating, and managing legal case information in the system.

**Audit Log.** Refers to the system-generated record of user actions and transactions, maintained to support traceability, accountability, and tamper-evident monitoring.

**Case Metadata.** Refers to the structured case-related information extracted or encoded in the system, such as case details, document attributes, and client-related data.

**Centralized Digital Case Archive.** Refers to the organized repository in JurisGuard where encoded and extracted legal case information is stored, managed, and retrieved securely.

**Cryptographic Protection.** Refers to the application of encryption and related security methods to protect legal records from unauthorized access, disclosure, or alteration.

**Data Integrity.** Refers to the accuracy, completeness, and consistency of legal case information stored and processed within the system.

**Document Management System (DMS).** Refers to the digital platform used to store, organize, manage, and secure documents and related records in a structured environment.

**Encryption at Rest.** Refers to the protection of stored data through encryption while the information resides in the database or storage system.

**Encryption in Transit.** Refers to the protection of data while it is being transmitted across networks to prevent interception or unauthorized access.

**Information Lifecycle Repository.** Refers to the system environment that manages legal case information from document intake and encoding up to storage, retrieval, updating, and case termination.

**JurisGuard.** Refers to the proposed intelligent legal document extraction and cryptographically secured archiving system developed for secure case information management.

**Legal Document Extraction.** Refers to the process of obtaining relevant case and client information from physical legal documents and converting it into structured digital data.

**Multi-Factor Authentication (MFA).** Refers to the authentication method that requires users to provide two or more verification factors before gaining access to the system.

**Optical Character Recognition (OCR).** Refers to the technology used by the system to recognize printed text from scanned or photographed legal documents and convert it into machine-readable form.

**Printed Documentation.** Refers to physical legal records containing printed English or Filipino text that are intended for OCR-assisted processing within the study.

**Role-Based Access Control (RBAC).** Refers to the access management approach in which permissions are assigned according to the specific roles and responsibilities of authorized users.

**Verification Interface.** Refers to the system feature that allows users to review, validate, and correct OCR-extracted case and client metadata before final storage.

**Zero Trust Architecture.** Refers to the security model that requires continuous verification of users and access requests rather than assuming trust based on network location or prior access.

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